

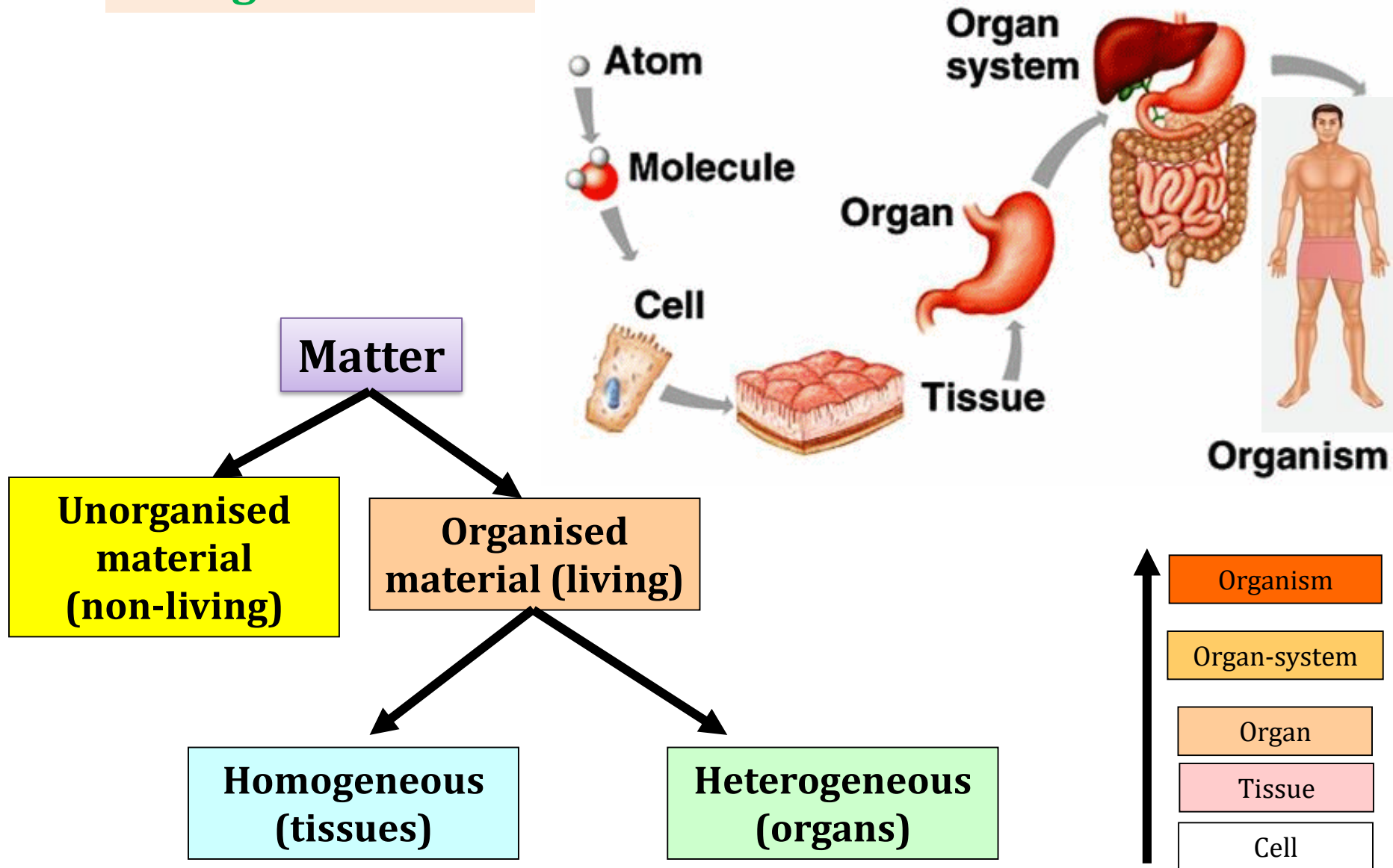
Cell structure and function

(Cellular foundation)

Lecture 11-13

Cells & Cell Organelles

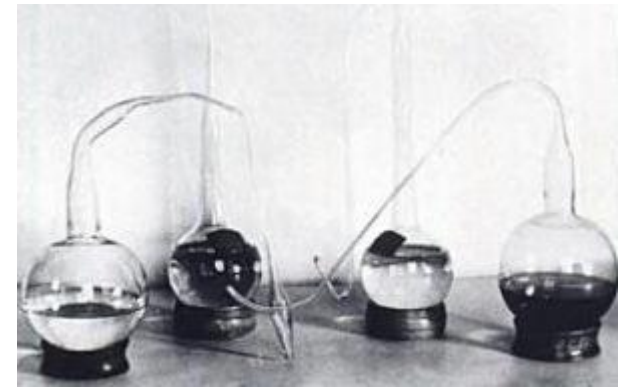
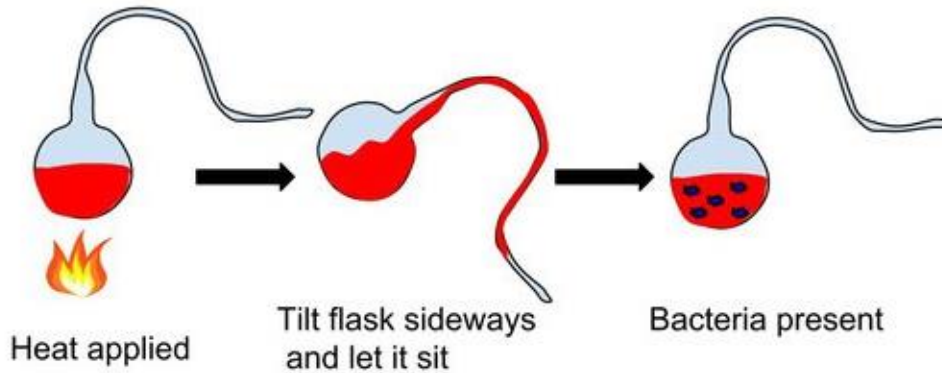
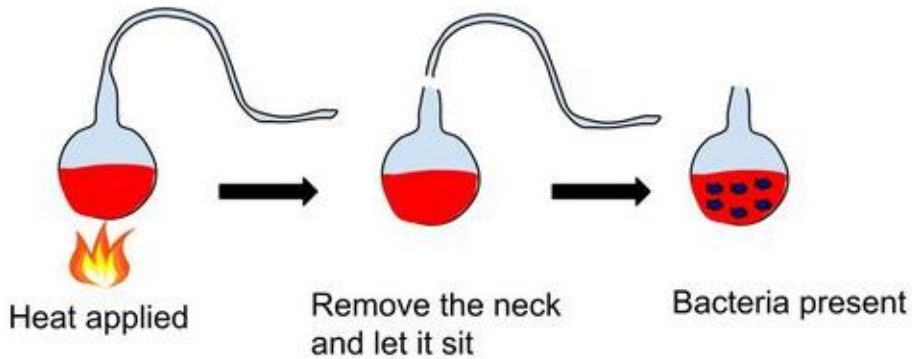
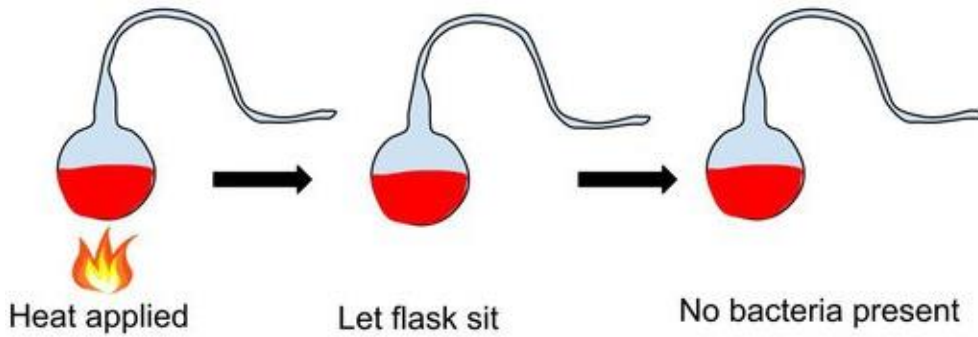
Doing Life's Work



Louis Pasteur – 1859/61

Pasteur's Swan-Neck Flask Experiment

[Law of Biogenesis]



Cell Theory

1. All organisms are made of 1 or more cells.

2. Cells are the basic building blocks of life.

3. All cells come from existing cells.

Rudolf Virchow

Theodore Shwann

Microscope

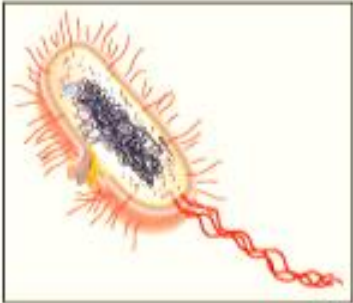


Theodore Shwann

Rudolf Virchow



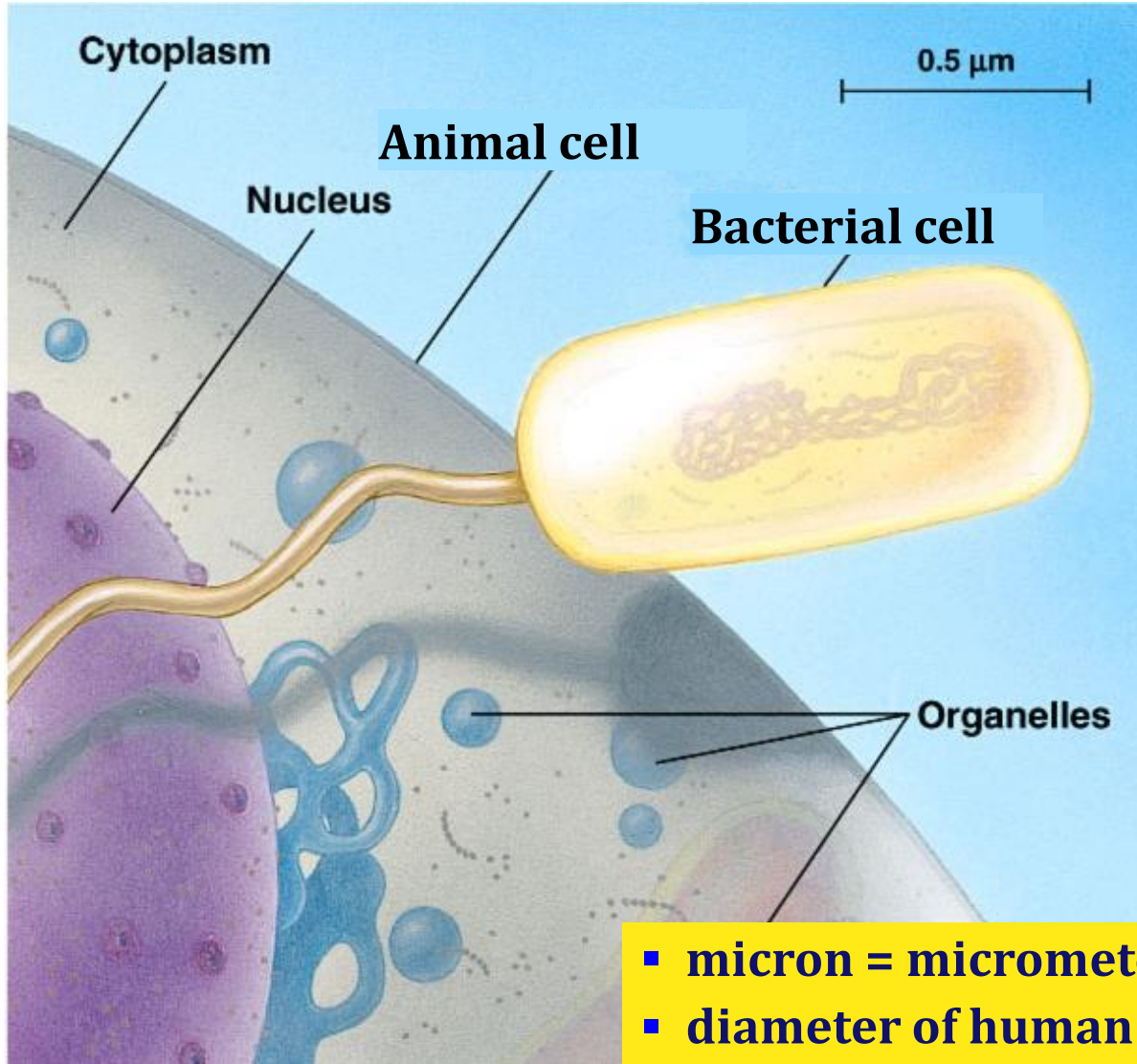
Principal Differences Between Prokaryotic and Eukaryotic Cells

Characteristic	Prokaryotic	Eukaryotic
		
Size of cell	Typically 0.2–2.0 μm in diameter	Typically 10–100 μm in diameter
Nucleus	No nuclear membrane or nucleoli	True nucleus, consisting of nuclear membrane and nucleoli
Membrane-enclosed organelles	Absent	Present; examples include lysosomes, Golgi complex, endoplasmic reticulum, mitochondria, and chloroplasts
Flagella	Consist of two protein building blocks	Complex; consist of multiple microtubules
Cell wall	Usually present; chemically complex (typical bacterial cell wall includes peptidoglycan)	When present, chemically simple
Plasma membrane	No carbohydrates and generally lacks sterols	Sterols and carbohydrates that serve as receptors present
Cytoplasm	No cytoskeleton or cytoplasmic streaming	Cytoskeleton; cytoplasmic streaming
Ribosomes	Smaller size (70S)	Larger size (80S); smaller size (70S) in organelles
Chromosome (DNA)	Single circular chromosome; lacks histones	Multiple linear chromosomes with histones arrangement
Cell division	Binary fission	Mitosis
Sexual reproduction	No meiosis; transfer of DNA fragments only	Involves meiosis

Animal and Plant cell comparison

	Animal Cell	Plant Cell
Cell wall	Absent	Present (formed of cellulose)
Shape	Round (irregular shape)	Rectangular (fixed shape)
Vacuole	One or more small vacuoles (much smaller than plant cells).	One, large central vacuole taking up 90% of cell volume.
Centrioles	Present in all animal cells	Only present in lower plant forms.
Chloroplast	Animal cells don't have chloroplasts	Plant cells have chloroplasts because they make their own food
Plastids	Absent	Present
Plasma Membrane	only cell membrane	cell wall and a cell membrane
Lysosomes	Lysosomes occur in cytoplasm.	Lysosomes usually not evident.
Cilia	Present	It is very rare

Cell size comparison



most bacteria

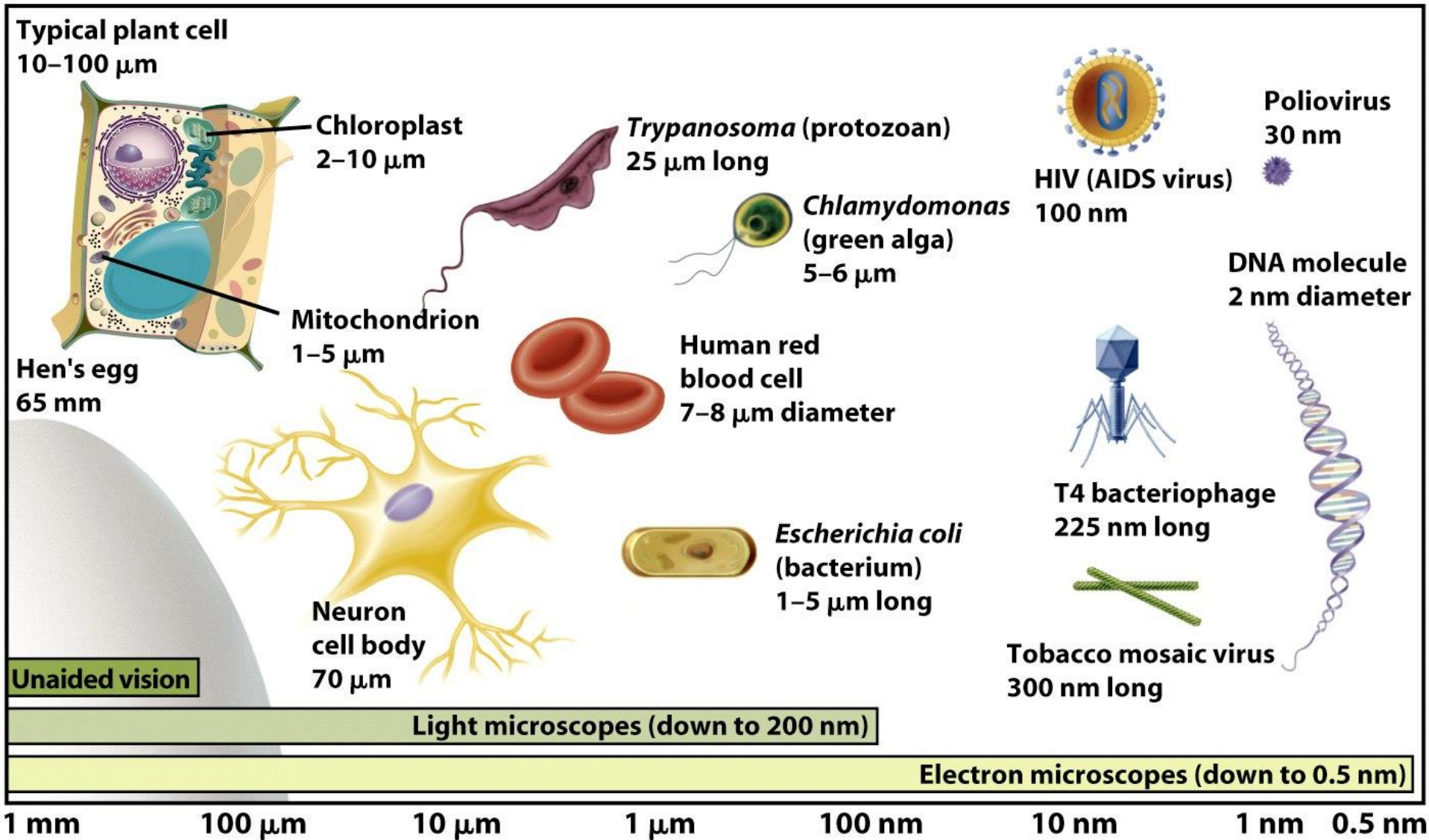
- 1-10 microns

eukaryotic cells

- 10-100 microns

- micron = micrometer = $1/1,000,000$ meter
- diameter of human hair = ~20 microns

Sizes of Living Things



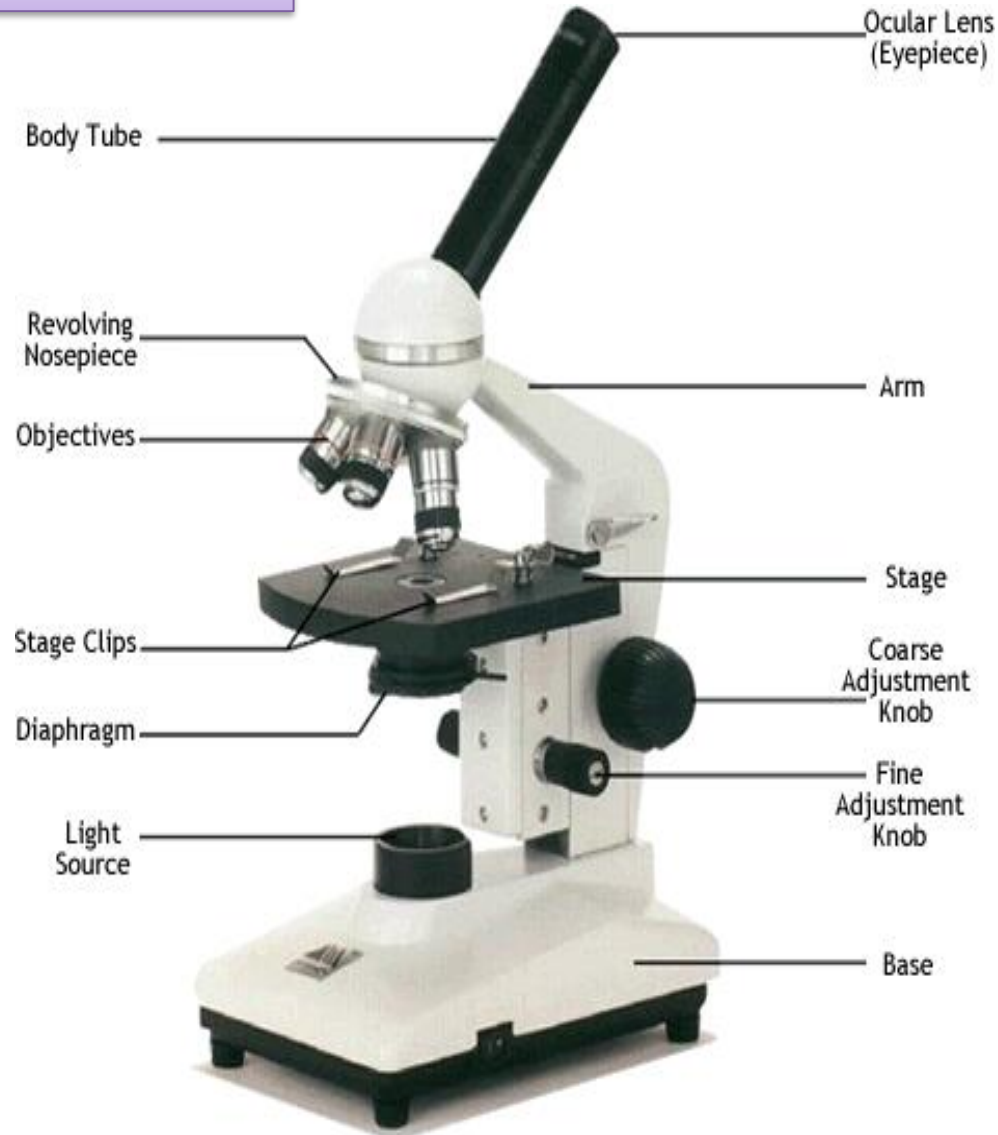
Problem: Cells are microscopic!

Solution: Use a microscope!

Types of Microscopes

Compound light microscope

- **Light passes through lenses to magnify image up to 1500X**
- **Can observe living cells**



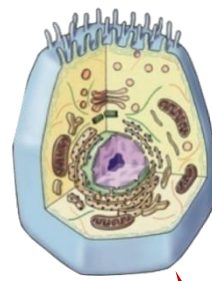
Types of Microscopes

Electron microscope

- Uses a beam of electrons to magnify image
> 1,000,000X
- Kills cells being observed



The Jobs of Cells



Our organelles
do all these
jobs!

- Cells have **3 main jobs**:

1. make energy

ATP

- need energy for all activities
- need to clean up waste produced while making energy

2. make proteins

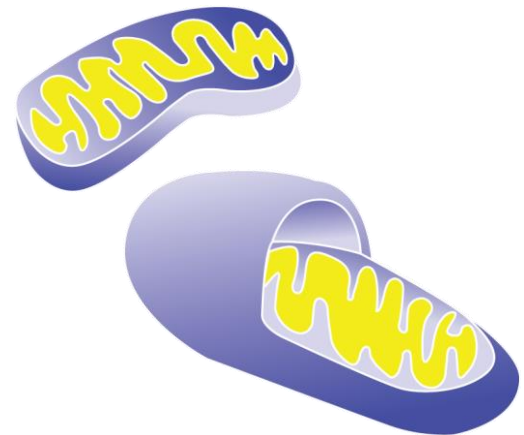
- proteins do all the work in a cell, so we need lots of them

3. make more cells

- for growth
- to replace damaged or diseased cells

1. Cells need power!

- Making energy
 - to fuel daily life & growth, the cell must...
 - take in food & digest it
 - take in oxygen (O_2)
 - make ATP
 - remove waste
 - organelles that do this work...
 - cell membrane
 - lysosomes
 - vacuoles & vesicles
 - mitochondria



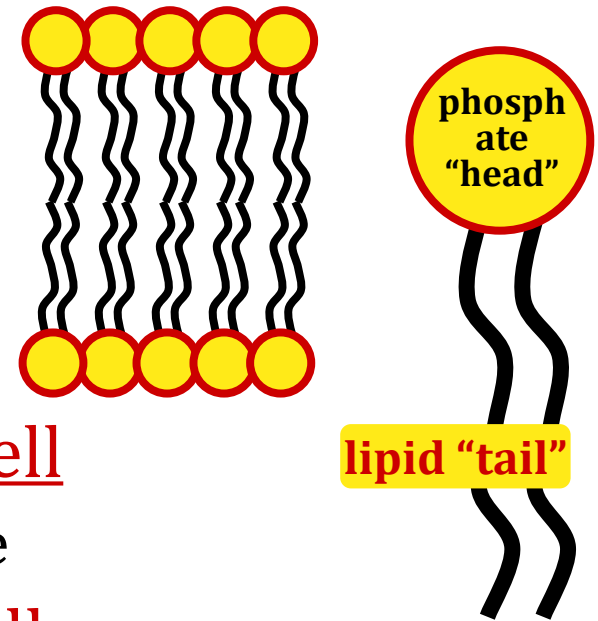
Cell membrane

- Function

- separates cell from outside
- controls what enters or leaves cell
 - O_2 , CO_2 , food, H_2O , nutrients, waste
- recognizes signals from other cells
 - allows communication between cells

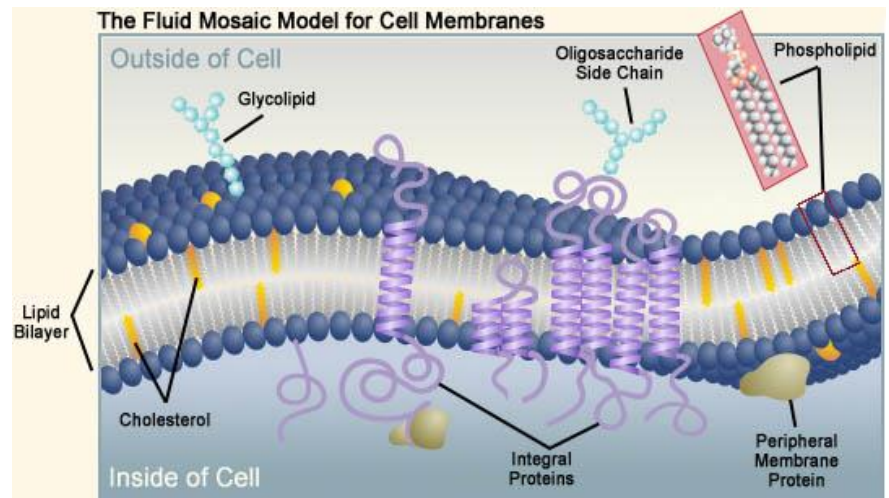
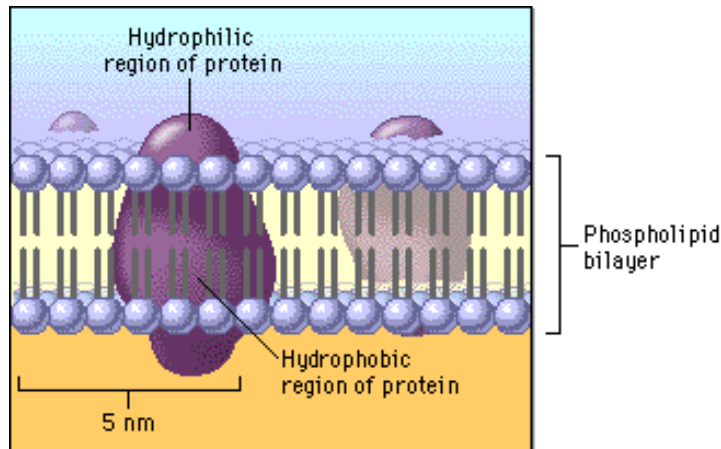
- Structure

- double layer of fat
 - phospholipid bilayer
- receptor molecules
 - proteins that receive signals



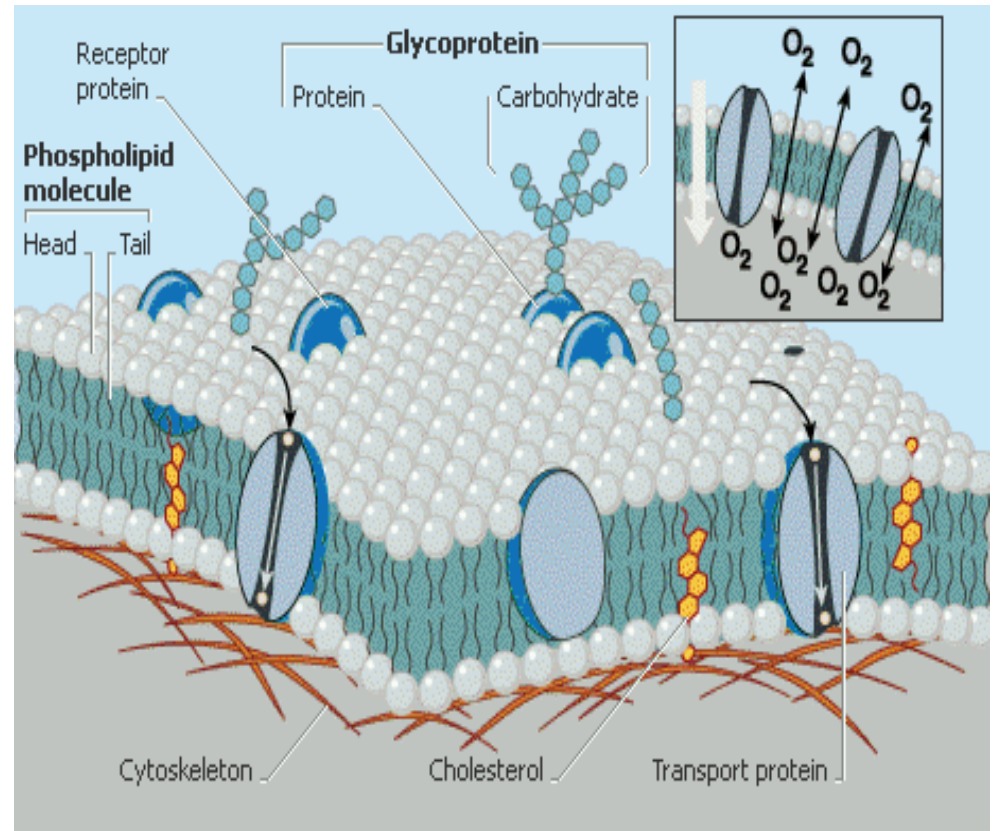
Fluid-Mosaic Model

- **Fluid** – the plasma membrane is the consistency of olive oil at body temperature, due to unsaturated phospholipids. (cells differ in the amount of unsaturated to saturated fatty acid tails)
- Most of the lipids and some proteins drift laterally on either side. Phospholipids do not switch from one layer to the next.
- Cholesterol affects fluidity: at body temperature, it lessens fluidity by restraining the movement of phospholipids, at colder temperatures it adds fluidity by not allowing phospholipids to pack close together.
- **Mosaic** – membrane proteins form a collage that differs on either side of the membrane and from cell to cell (greater than 50 types of proteins), proteins span the membrane with hydrophilic portions facing out and hydrophobic portions facing in. Provides the functions of the membrane



How do materials move into and out of the cell?

- Materials must move in and out of the cell through the plasma membrane.
- Some materials move **between the phospholipids**.
- Some materials move **through the proteins**.



Plasma Membrane Transport

Passive Transport

Active Transport

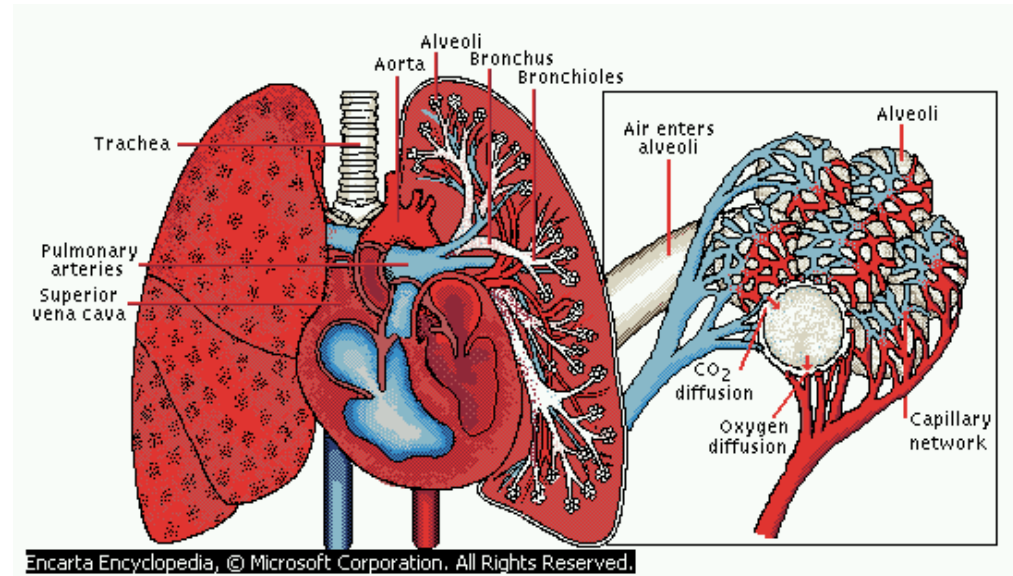
Passive Transport:

ATP energy is **not** needed to move the molecules through.

1. Molecules can move directly through the phospholipids of the plasma membrane – **Diffusion**
2. Molecules can move through the plasma membrane with the aid of transport proteins - **Facilitated Diffusion**
3. Water Molecules can move directly through the phospholipids of the plasma membrane - **Osmosis**

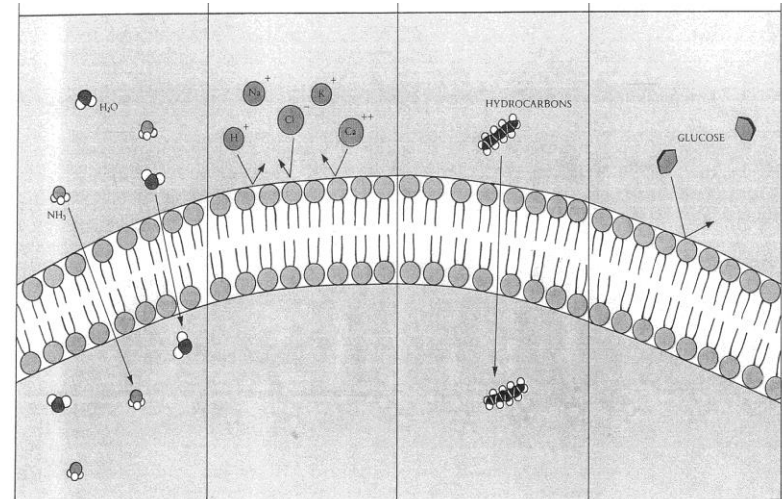
Why is diffusion important to cells and humans?

- Cell respiration
- Alveoli of lungs
- Capillaries
- Red Blood Cells
- Medications: time-release capsules



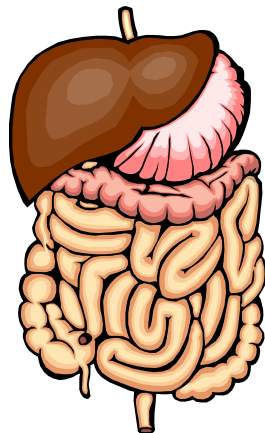
What molecules move through the plasma membrane by **facilitated diffusion**?

- Ions (Na^+ , K^+ , Cl^-)
- Sugars (Glucose)
- Amino Acids
- Small water soluble molecules
- Water (faster rate)



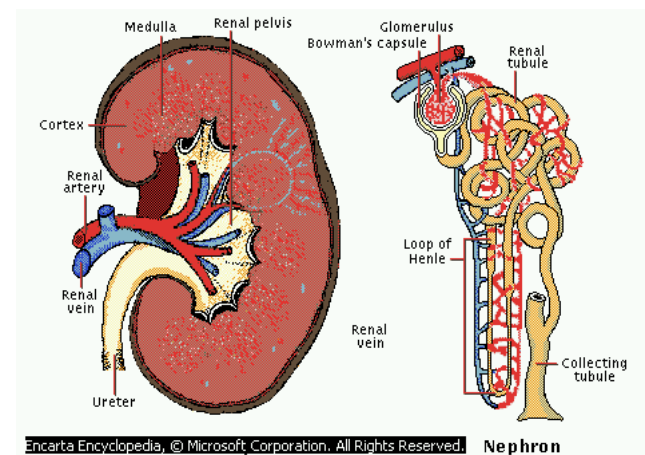
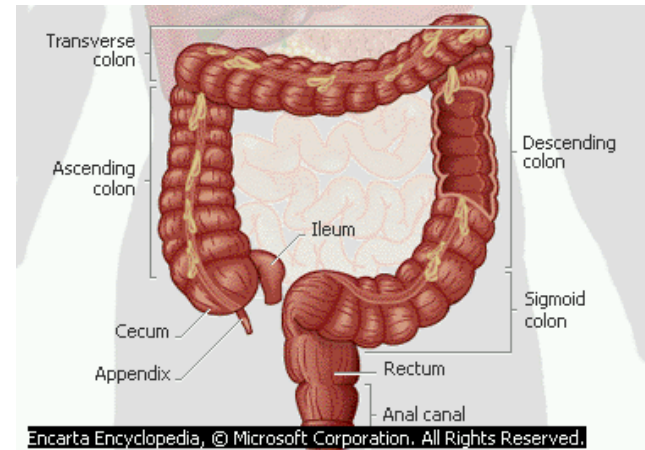
Why is **facilitated diffusion** important to cells and humans?

- Cells obtain food for cell respiration
- Neurons communicate
- Small intestine cells **transport food to the bloodstream**
- Muscle cells contract



Why is **osmosis** important to cells and humans?

- Cells **remove water** produced by cell respiration.
- Large intestine cells **transport water** to the bloodstream
- Kidney cells form **urine**



What are three types of Active transport?

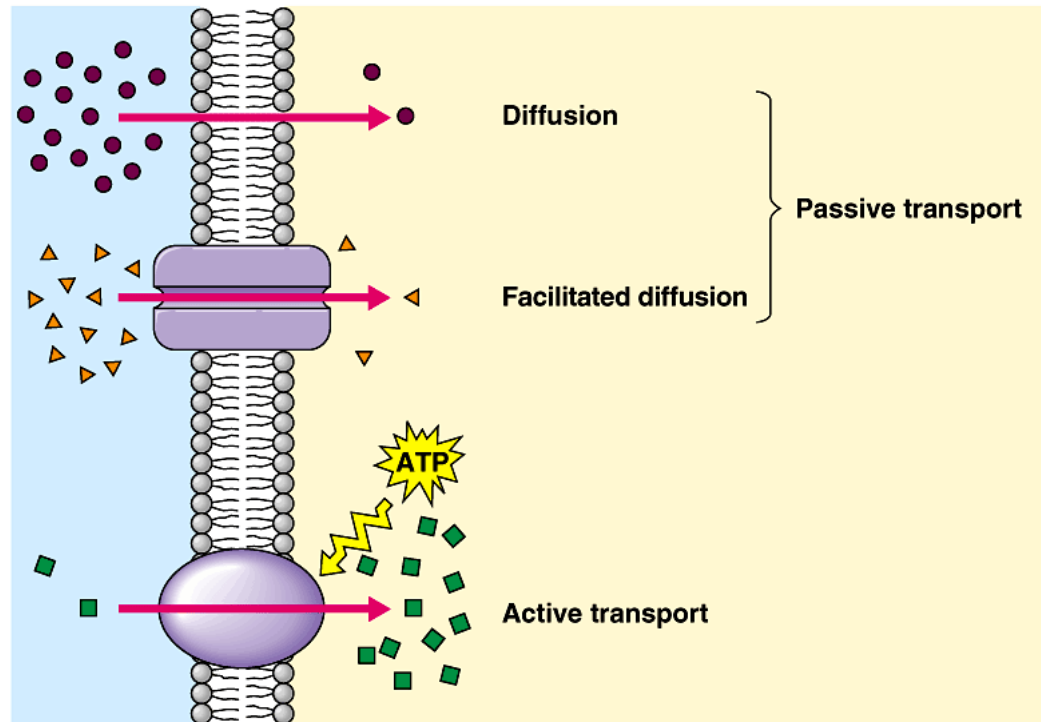
Active Transport

- 1) Active Transport
- 2) Exocytosis
- 3) Endocytosis
 - Phagocytosis
 - Pinocytosis
 - Receptor-Mediated endocytosis

ATP energy is **required** to move the molecules through.

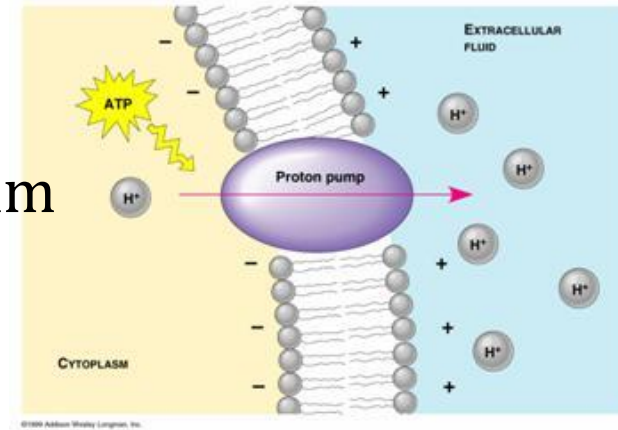
Active Transport

- Molecules move from areas of low concentration to areas of high concentration with the aid of ATP energy.
- Requires protein carriers called **Pumps**.



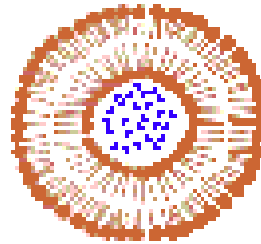
The Importance of Active Transport

- Bring in essential molecules: **ions, amino acids, glucose, nucleotides**
- Rid cell of unwanted molecules (Ex. sodium from urine in kidneys)
- Maintain internal conditions different from the environment
- **Regulate the volume of cells** by controlling osmotic potential
- Control cellular pH
- **Re-establish concentration gradients** to run facilitated diffusion. (Ex. Sodium-Potassium pump and Proton pumps)



Active Transport 2: Exocytosis

- Movement of large molecules bound in vesicles **out of** the cell with **the aid of ATP energy**. Vesicle fuses with the plasma membrane to eject macromolecules.
- Export Proteins, polysaccharides, polynucleotides, whole cells, hormones, mucus neurotransmitters, waste



VESICLE



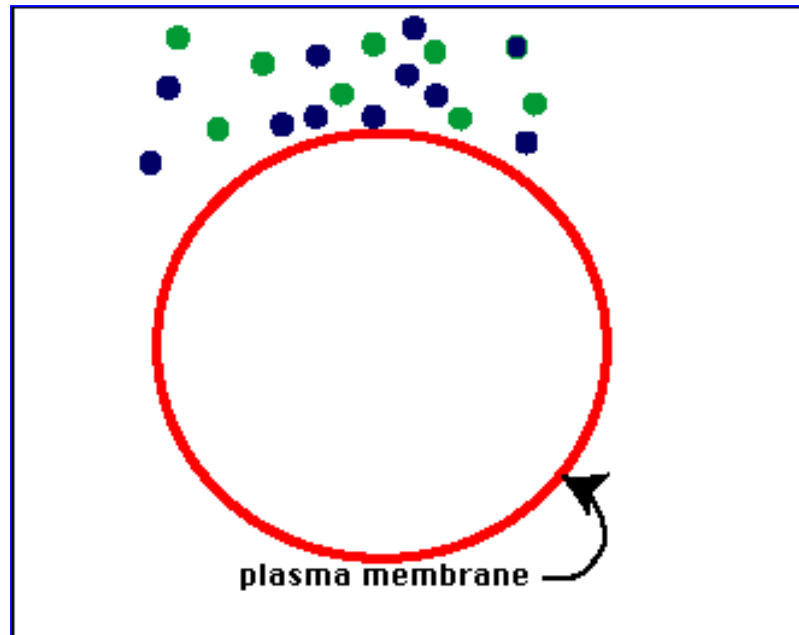
PLASMA
MEMBRANE

Active Transport 3: Endocytosis

- Movement of large molecules **into** the cell by engulfing them in vesicles, using ATP energy.
- Three types of Endocytosis:
 - a) Phagocytosis
 - b) Pinocytosis
 - c) Receptor-mediated endocytosis

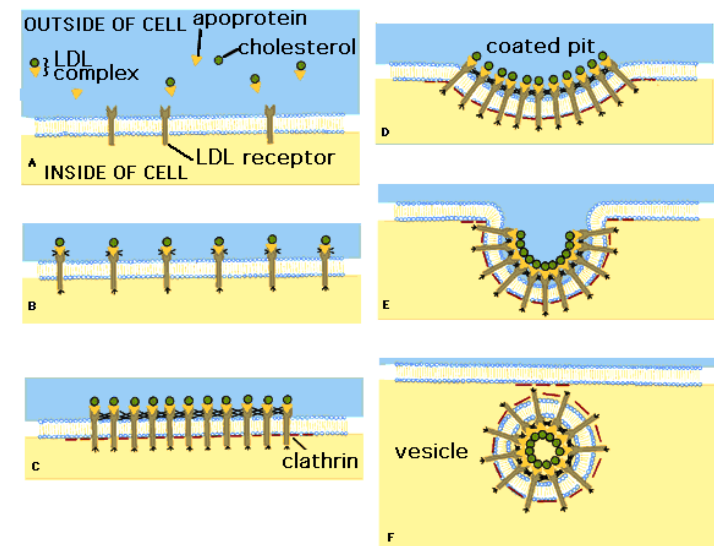
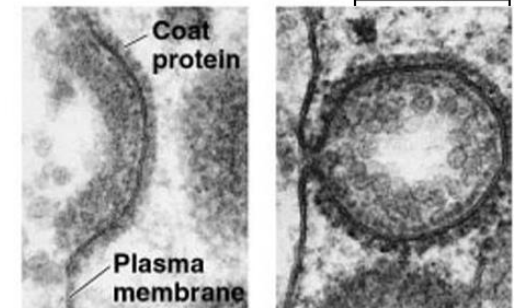
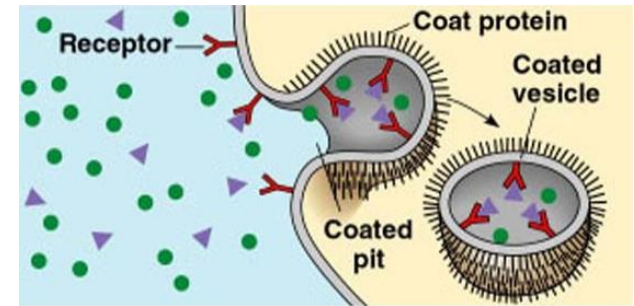
b) Pinocytosis

- “Cellular Drinking” – engulfing liquids and **small molecules dissolved in liquids**; unspecific what enters.
- Ex. Intestinal cells, Kidney cells, Plant root cells

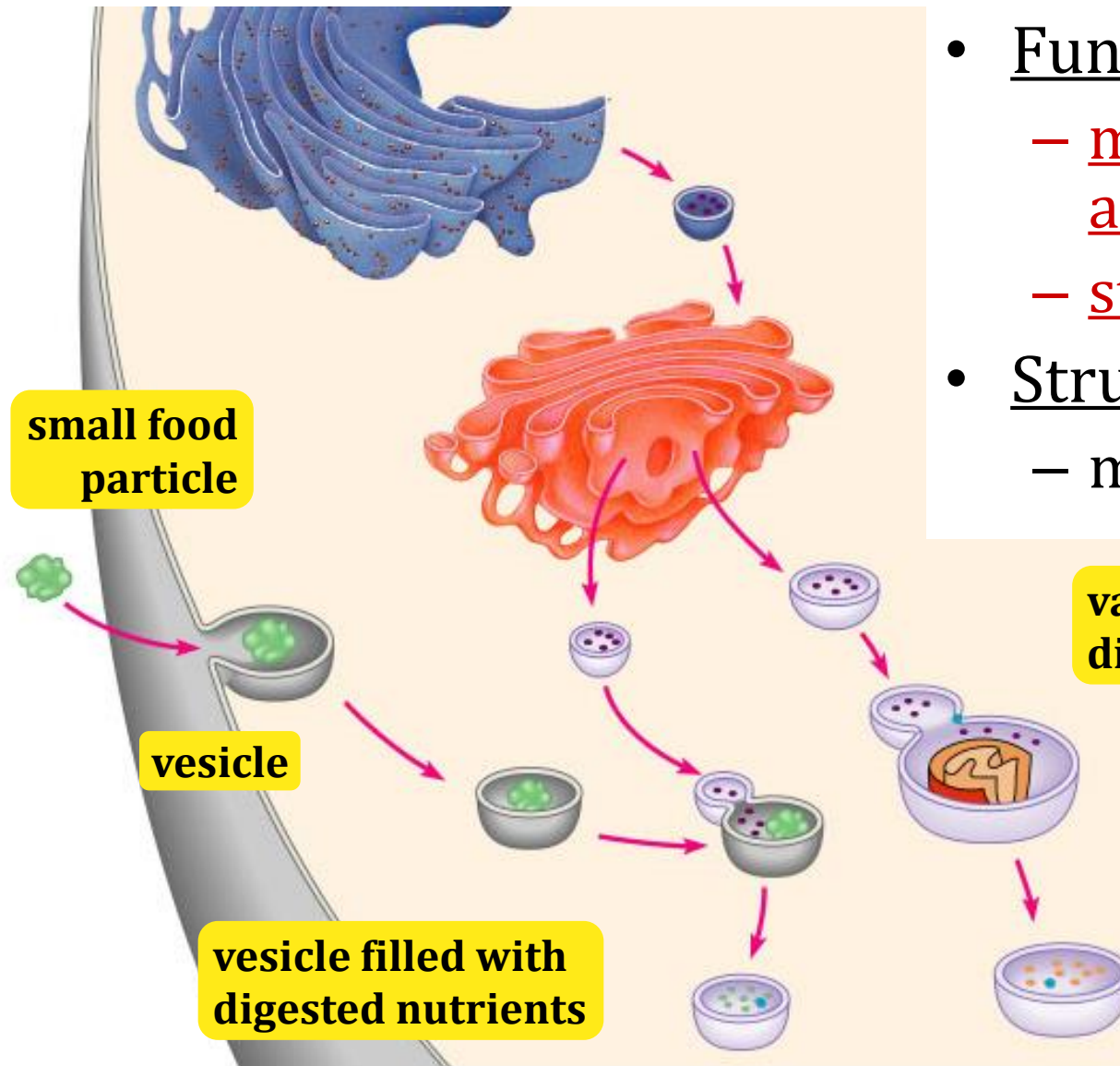


c) Receptor-Mediated Endocytosis

- Movement of very specific molecules into the cell with the use of vesicles coated with the protein clathrin.
- Coated pits are specific locations coated with clathrin and receptors. When specific molecules (**ligands**) bind to the receptors, then this stimulates the molecules to be engulfed into a coated vesicle.
- Ex. Uptake of cholesterol (LDL) by animal cells



Vacuoles & vesicles

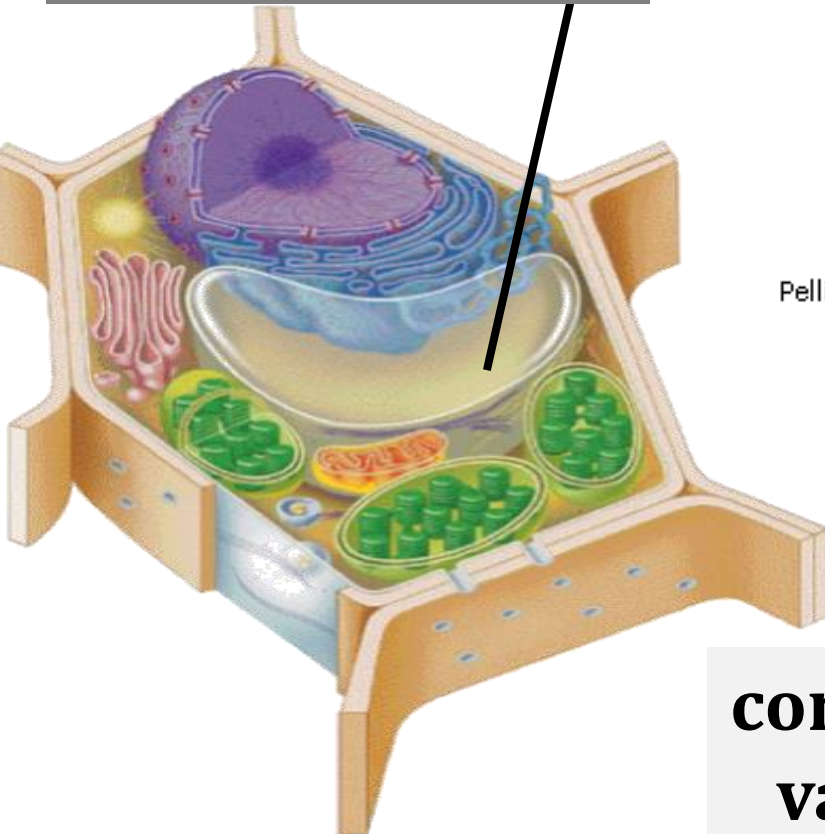


- Function
 - moving material around cell
 - storage
- Structure
 - membrane sac

Food & water storage

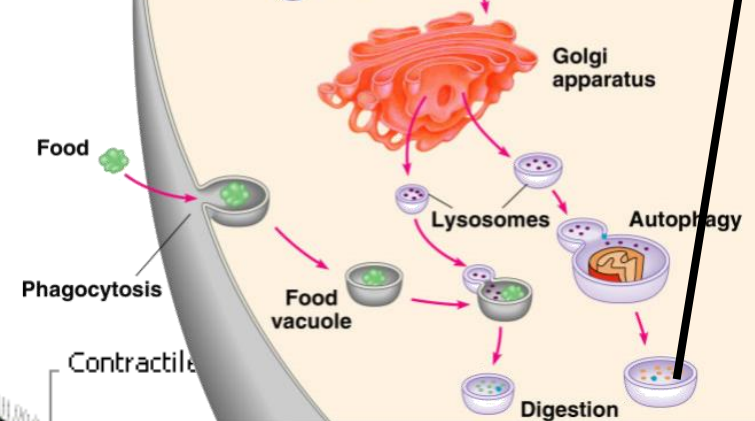
plant cells

central vacuole



Plasma membrane

food vacuole



Food vacuole

Contractile

Pellicle

Cilia

Cytostome

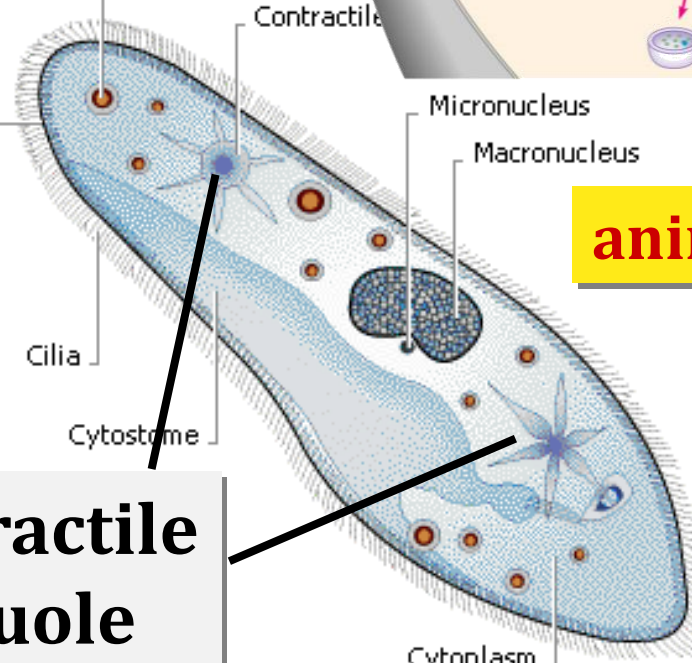
contractile vacuole

Micronucleus

Macronucleus

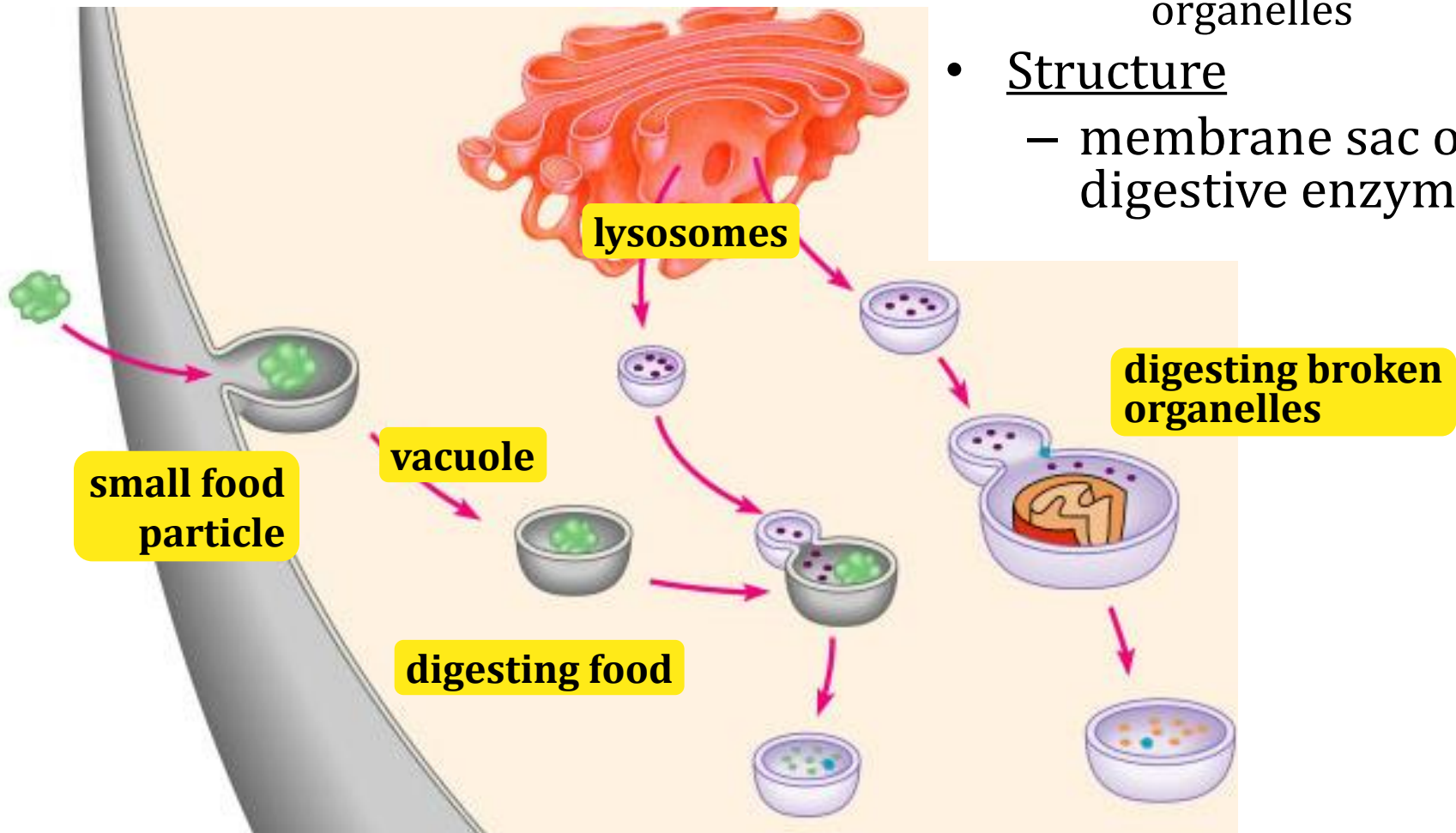
animal cells

Cytoplasm



Lysosomes

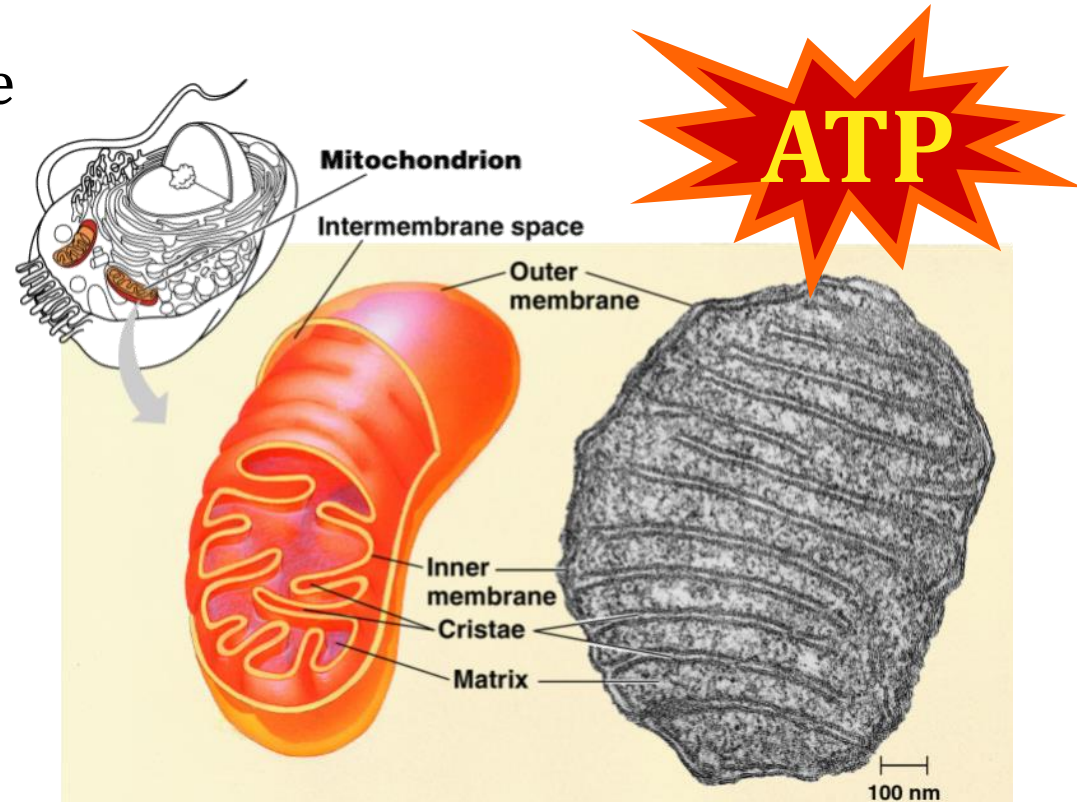
- Function
 - digest food
 - used to make energy
 - clean up & recycle
 - digest broken organelles
- Structure
 - membrane sac of digestive enzymes



Mitochondria

- Function:
 - make ATP energy from cellular respiration
 - $\text{sugar} + \text{O}_2 \rightarrow \text{ATP}$
 - fuels the work of life
- Structure
 - double membrane

in both animal & plant cells



cytoplasm

- jelly-like material holding organelles in place

vacuole & vesicles

- transport inside cells
- storage

lysosome

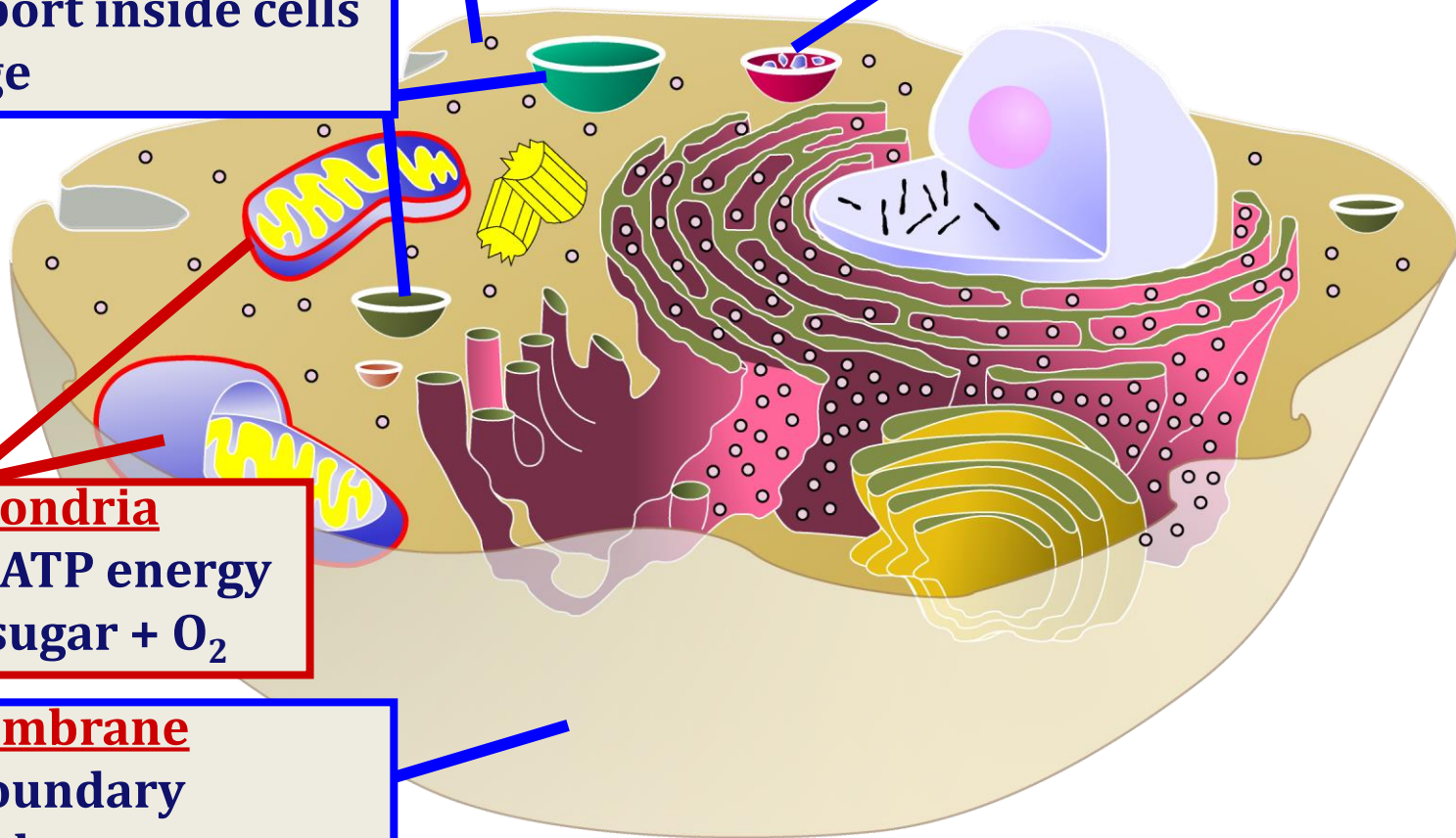
- food digestion
- garbage disposal & recycling

mitochondria

- make ATP energy from sugar + O₂

cell membrane

- cell boundary
- controls movement of materials in & out
- recognizes signals



Plants make energy two ways!

1. Mitochondria

- make energy from sugar + O_2
 - cellular respiration
 - $\text{sugar} + O_2 \rightarrow \text{ATP}$



2. Chloroplasts

- make energy + sugar from sunlight
 - photosynthesis
 - $\text{sunlight} + CO_2 \rightarrow \text{ATP \& sugar (Starch)}$
 - ATP = active energy
 - Sugar (starch) = stored energy
 - » build leaves & roots & fruit out of the sugars

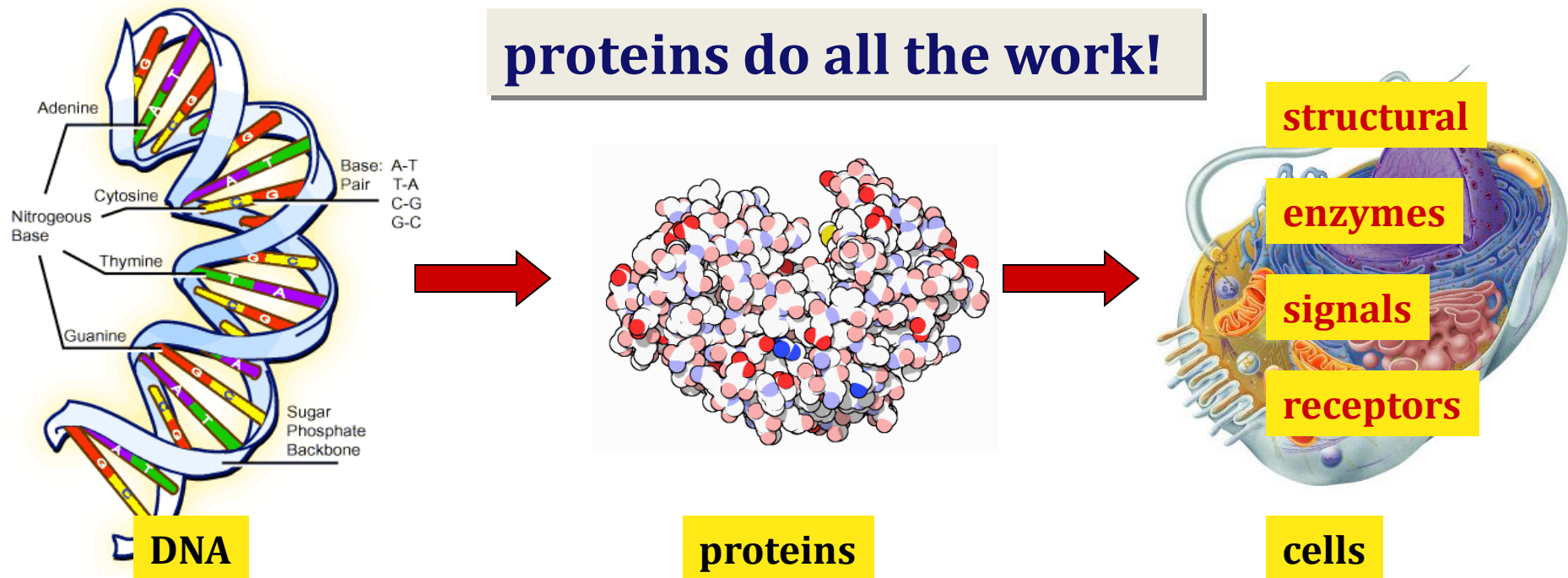


2. Cells need workers = proteins!

- Making proteins
 - to run daily life & growth, the cell must...
 - read genes (DNA)
 - build proteins
 - structural proteins (muscle fibers, hair, skin, claws)
 - enzymes (speed up chemical reactions)
 - signals (hormones) & receptors
 - organelles that do this work...
 1. nucleus
 2. ribosomes
 3. endoplasmic reticulum (ER)
 4. golgi apparatus

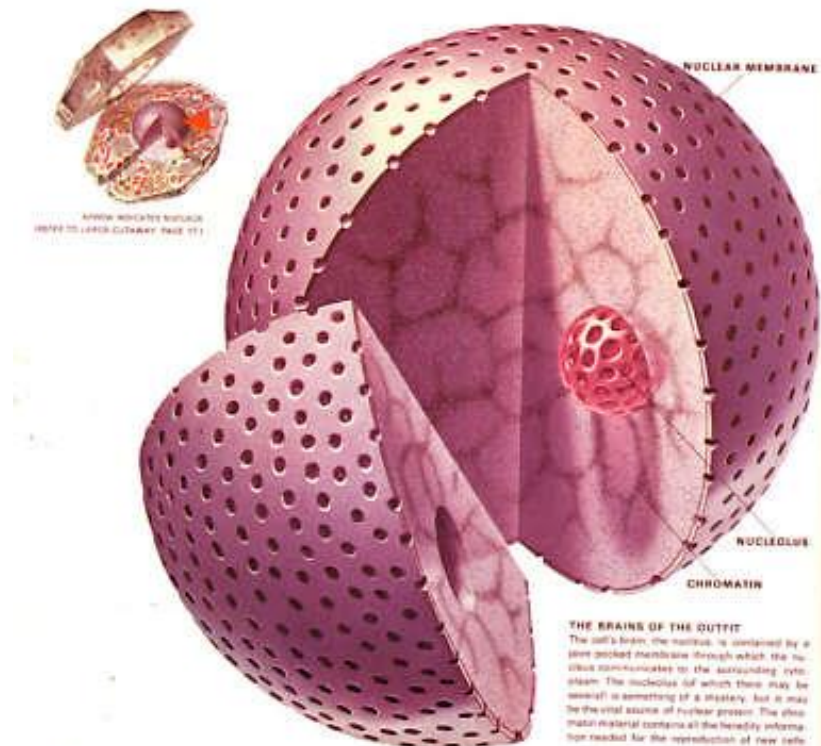
Proteins do all the work!

one of the major job of cells is to make proteins,
because...



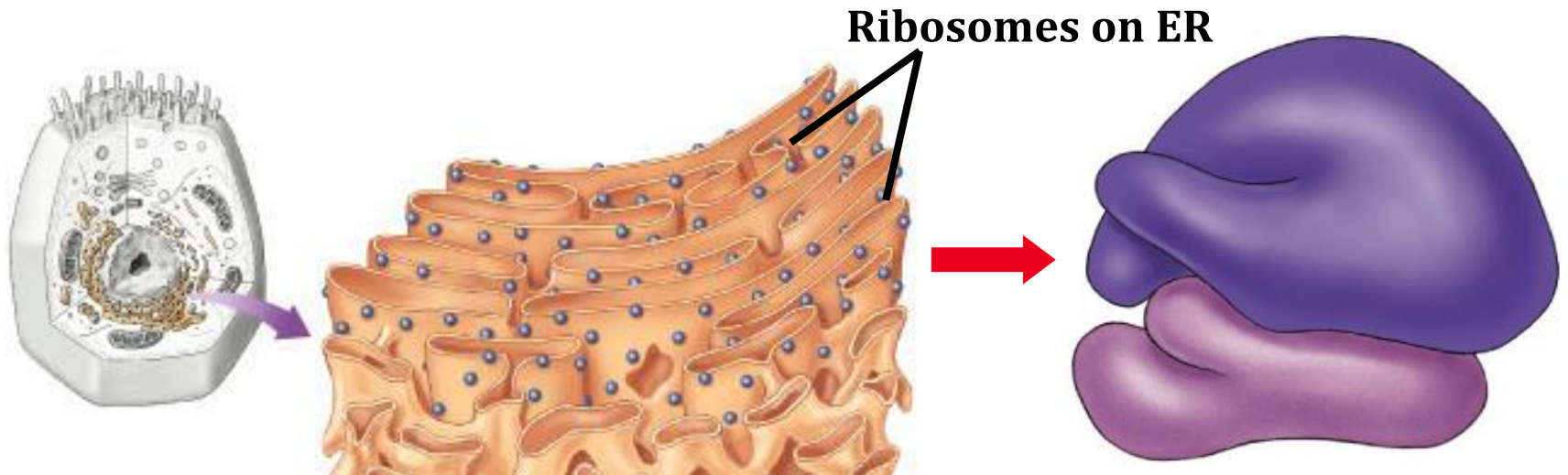
Nucleus

- Function
 - control center of cell
 - protects DNA
 - instructions for building proteins
- Structure
 - nuclear membrane
 - nucleolus
 - ribosome factory
 - chromosomes
 - DNA



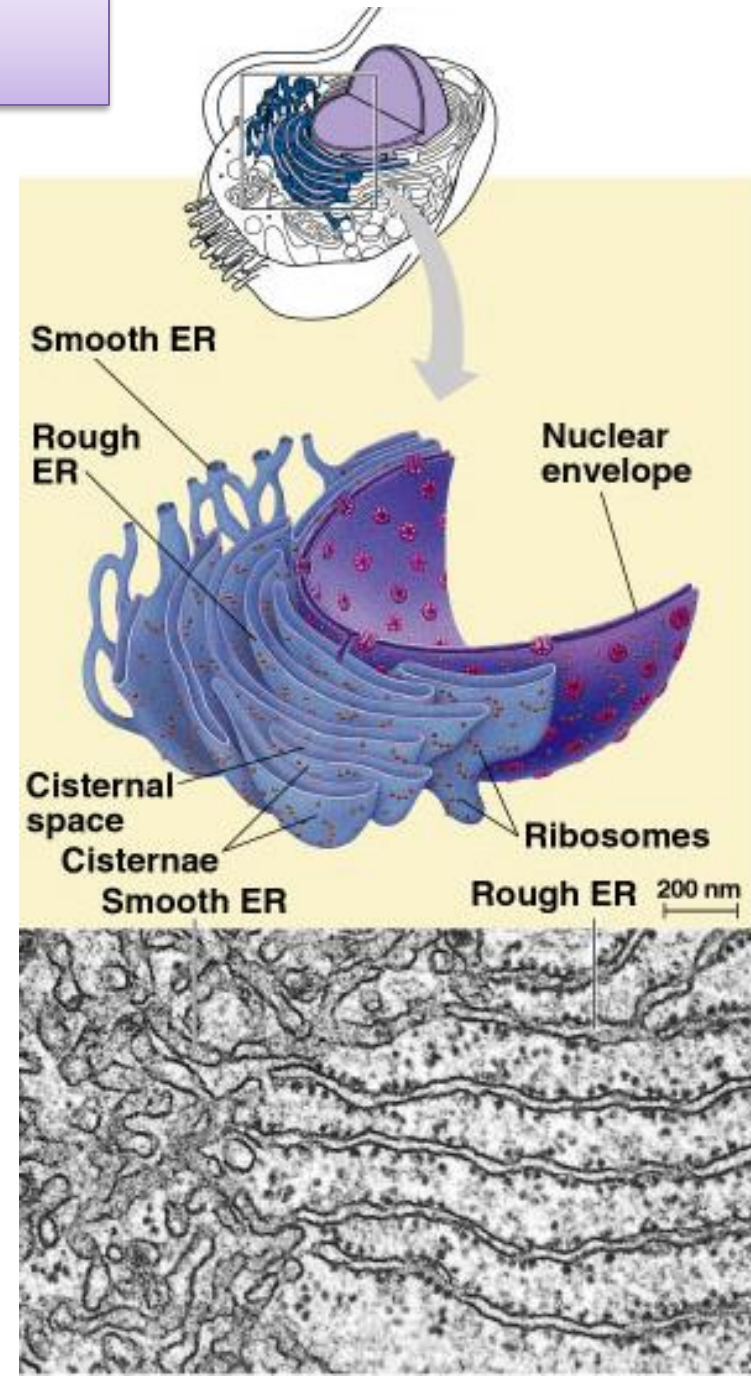
Ribosomes

- Function
 - protein factories
 - read instructions to build proteins from DNA
- Structure
 - some free in cytoplasm
 - some attached to ER



Endoplasmic Reticulum

- Function:
 - works on proteins
 - helps complete the proteins after ribosome builds them
 - makes membranes
- Structure
 - rough ER
 - ribosomes attached
 - works on proteins
 - smooth ER
 - makes membranes
 - Detoxify drugs and toxins



Golgi Apparatus

- Function

- finishes, sorts, labels & ships proteins

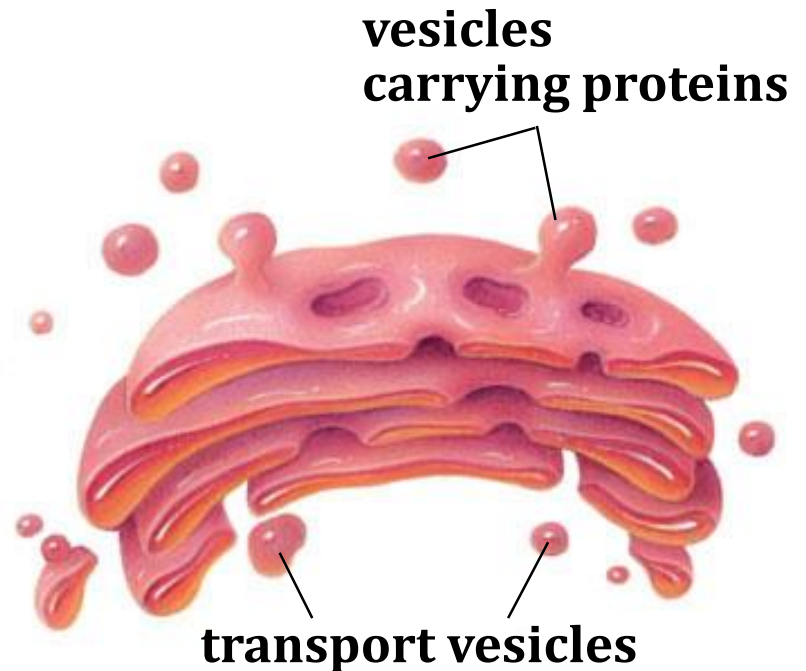
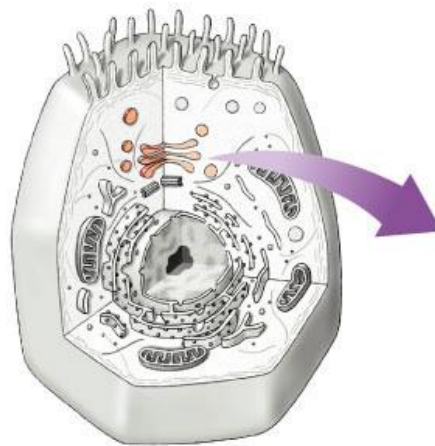
- like UPS headquarters
 - shipping & receiving department

- ships proteins in vesicles

- “UPS trucks”

- Structure

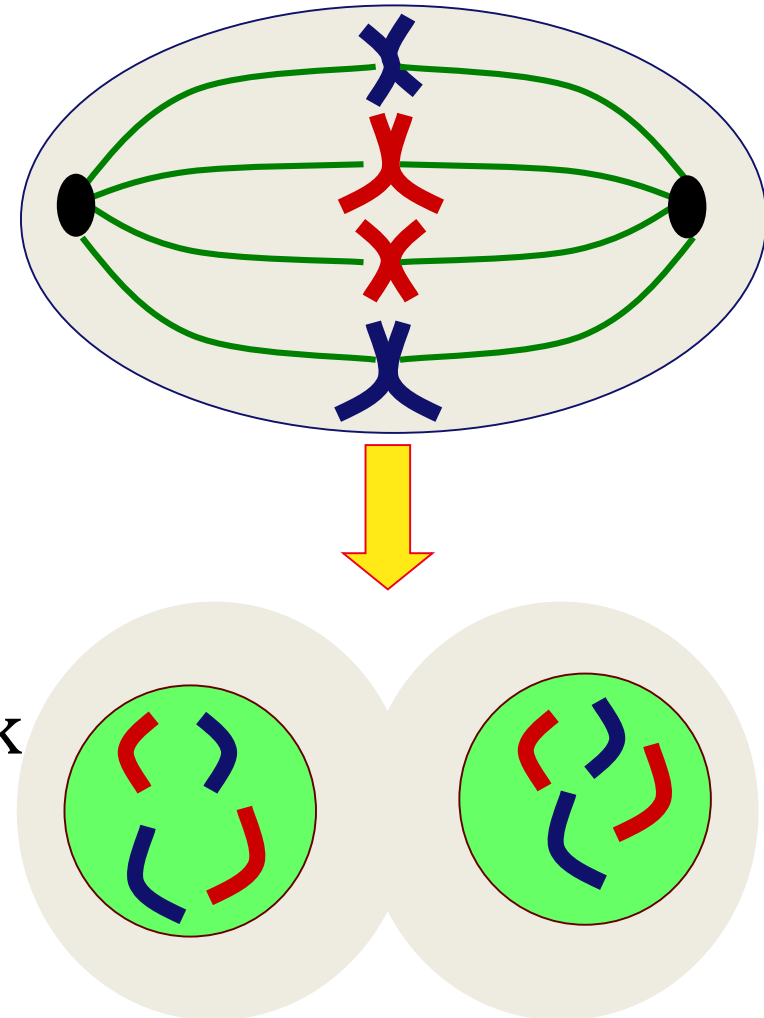
- membrane sacs



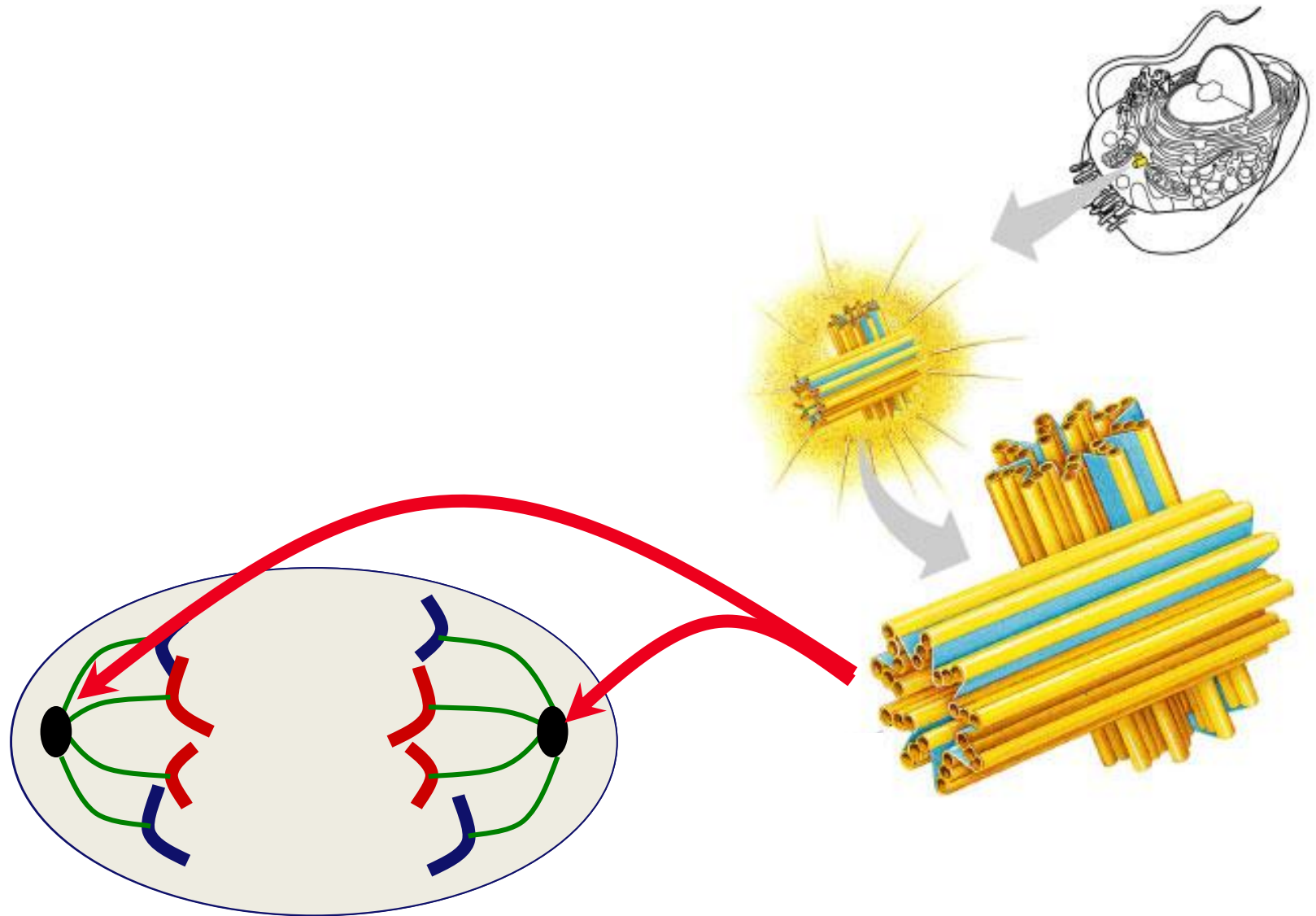
3. Cells need to make more cells!

- Making more cells

- to replace, repair & grow, the cell must...
 - copy their DNA
 - make extra organelles
 - divide the new DNA & new organelles between 2 new “daughter” cells
- organelles that do this work
 - nucleus
 - centrioles



Centrioles



cytoplasm

- jelly-like material holding organelles in place

vacuole & vesicles

- transport inside cells
- storage

lysosome

- food digestion
- garbage disposal & recycling

nucleus

- protects DNA
- controls cell

centrioles

- cell division

mitochondria

- make ATP energy from sugar + O₂

cell membrane

- cell boundary
- controls movement of materials in & out
- recognizes signals

ER

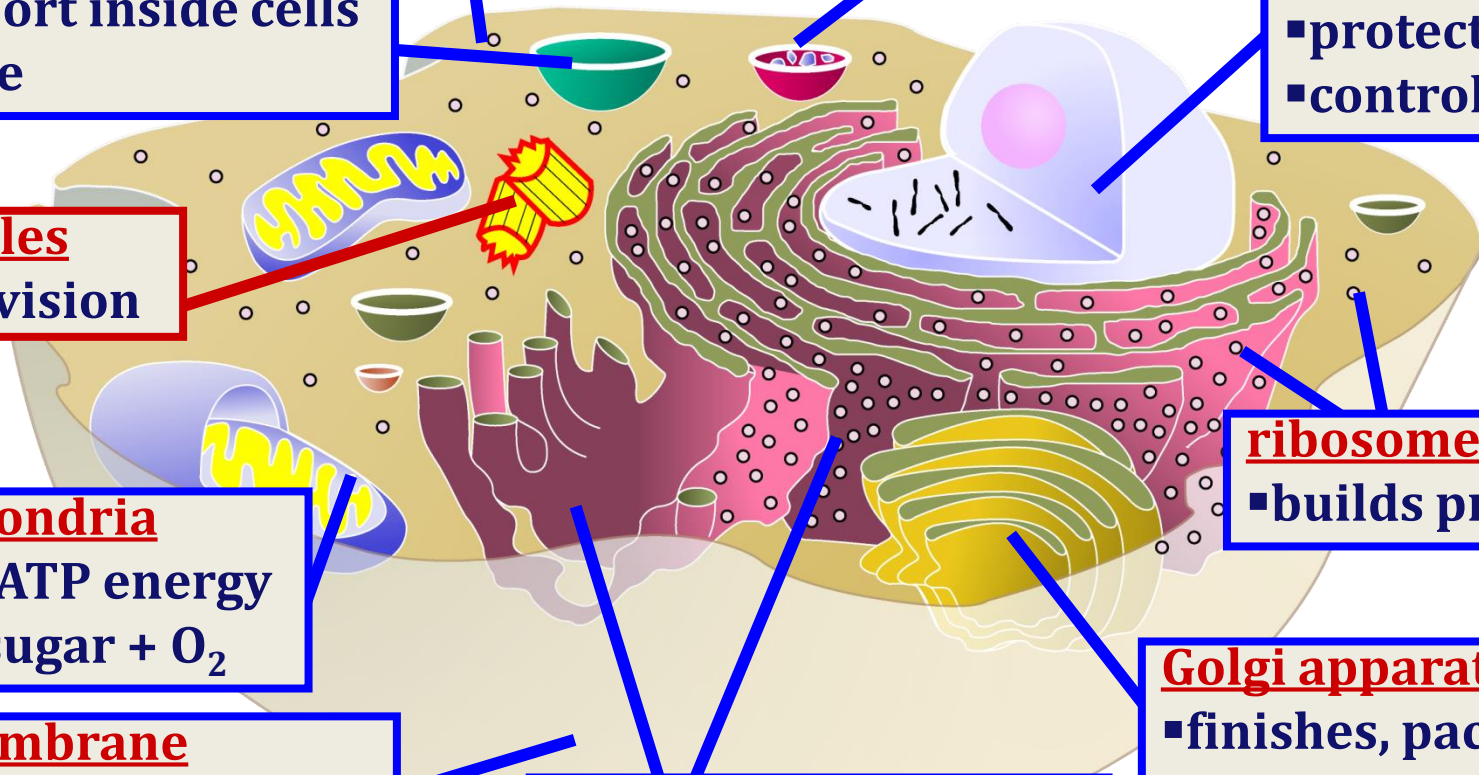
- helps finish proteins
- makes membranes

ribosomes

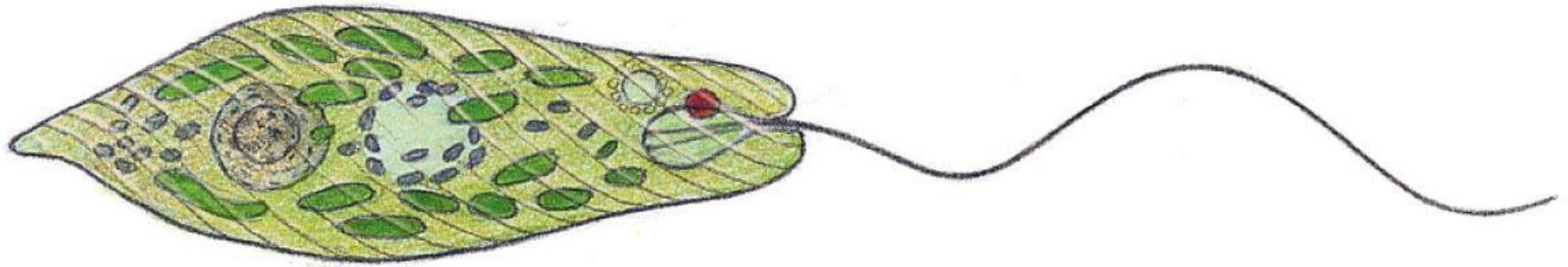
- builds proteins

Golgi apparatus

- finishes, packages & ships proteins

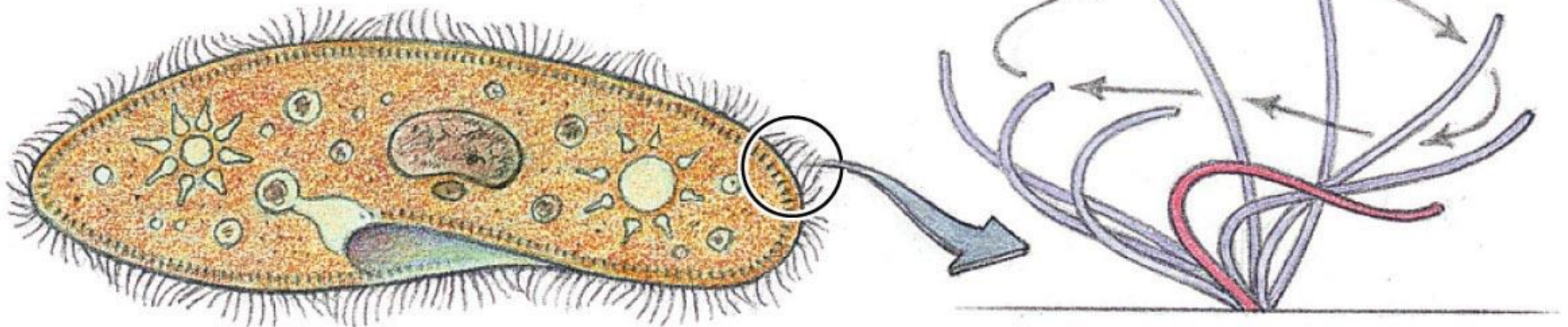


Direction of motion



(a) Flagella

Direction of motion



(b) Cilia

